Outline: Sentencing Disparities between Sexes

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Introduction and Motivation

- Investigating claims on disparities between sexes for sentencing is important because it highlights potential flaws in our justice system.
- The justice system should operate under the notion that everyone is innocent until proven guilty.
- All individuals deserve a fair trial and equitable treatment.
- Public trust depends on the court's commitment to due diligence and justice for all.

Project Goal

- Investigate the claim that women are sentenced more harshly than men for homicide cases in Florida.
- Develop a scalable, automated framework.
- Framework should be applicable to other types of offenses.

Literature Review and Institutional Details

- Review of prior studies on judicial bias—mostly race-based.
- Overview of economic and statistical theories of discrimination.
- These works provide conceptual and methodological foundations for analyzing sex-based disparities.

A Test of Racial Bias in Capital Sentencing

Authors: Alberto Alesina and Eliana La Ferrara
Publication: American Economic Review, 2014

- Examines racial bias via sentence reversal patterns.
- Finds higher reversal rates for minority defendants who kill white victims.
- Suggests unequal treatment, especially in Southern states.
- **Relevance**: Informs modeling of judicial decisions and subgroup analysis.

A Review of Thomas Sowell's *Discrimination and Disparities*

Author: Jennifer L. Doleac
Publication: Journal of Economic Literature, 2021

- Reviews Sowell's view: disparities stem from many sources beyond discrimination.
- Emphasizes culture, choices, and complexity of inequality.
- **Relevance**: Encourages cautious interpretation of observed disparities.

The Economics of Discrimination

Author: Gary S. Becker
Publication: University of Chicago Press, 1957

- Introduces taste-based discrimination.
- Prejudice leads to inefficient market outcomes.
- **Relevance**: Judges may act on personal biases even against rational incentives.

The Theory of Discrimination

Author: Kenneth Arrow
Publication: Industrial Relations Section, Princeton, 1971

- Introduces statistical discrimination model.
- Group characteristics used as proxies due to imperfect info.
- **Relevance**: Helps frame sex as a proxy variable in sentencing outcomes.

The Economics of Discrimination: Economists Enter the Courtroom

Authors: Orley Ashenfelter and Ronald Oaxaca
Publication: American Economic Review, 1987

- Explains Oaxaca–Blinder decomposition in civil discrimination cases.
- Outcome gaps separated into explained and unexplained components.
- **Relevance**: Same method can be adapted to criminal sentencing analysis.

Theoretical Model: Sentencing as Statistical Decision

- Judges modeled as decision-makers minimizing sentencing errors:
  - **Type I Error**: Over-penalizing (harsh sentence)
  - **Type II Error**: Under-penalizing (too lenient, public risk)
- Inspired by Alesina and La Ferrara (2014)
- In absence of bias, sentencing should depend only on:
  - Charge type, prior record, plea status, etc.
- Disparities conditional on these imply bias in inferred culpability.

Sentence as Maximum Likelihood Estimate

MLE of latent culpability/dangerousness, constrained by statutory bounds.

$$y_i = \frac{\log(\mathrm{Sentence}_i)}{\log(\mathrm{Max}_i) - \log(\mathrm{Min}_i)}, \quad \mathrm{Min}_i > 0$$

- $y_i \in [0, 1]$: scaled sentence severity
- Captures proportional judicial discretion
- Adjusts for skew in sentence distribution

Logarithmic Sentencing Ratio Visualization

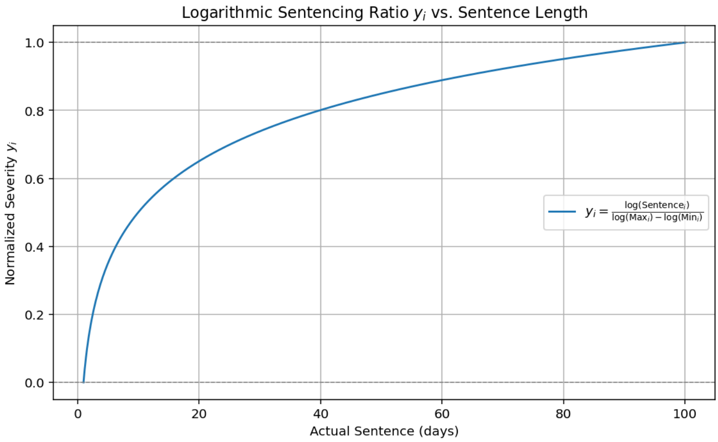

Illustration of normalized sentencing severity y_i as a function of actual sentence length.

Life Sentences as Right-Tail Point Mass

- Life sentences modeled separately—qualitatively different.
- Treated as degenerate distribution (extreme severity).

$$\Pr(\text{Life}_i = 1) = \Phi(\alpha_0 + \alpha_1 S_i + \alpha_2 X_i)$$

- S_i: sex
- X_i: control variables
- Captures likelihood of life sentencing as a binary probit model

Sentencing Severity Distribution and Life Sentences

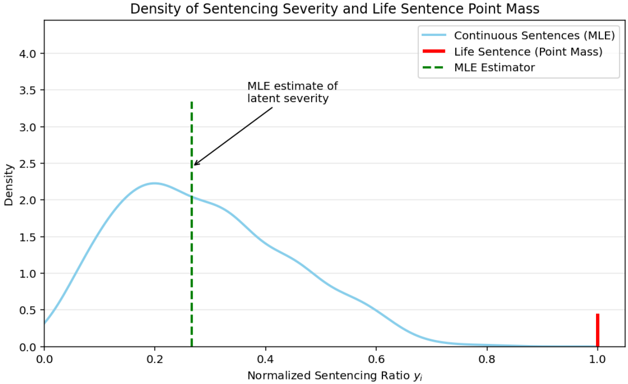

Figura 1: Density of normalized sentencing ratios y_i with a point mass at 1 representing life sentences.

Description:

- The smooth curve represents the density of continuous sentencing

Economic Discrimination Theory

Two frameworks for judicial bias:

- **Taste-Based Discrimination** (Becker, 1957)
  - Judges act on personal preferences/stereotypes
- **Statistical Discrimination** (Arrow, 1971)
  - Judges use sex as proxy for unobservables (e.g., risk)
- Both mechanisms produce disparities not explained by case facts

Oaxaca–Blinder Decomposition (Log Outcomes)

Goal: Decompose gender gap in sentencing severity.

$$\log \bar{y}_M - \log \bar{y}_F = (\bar{X}_M - \bar{X}_F)\hat{\beta}_M + \bar{X}_F(\hat{\beta}_M - \hat{\beta}_F)$$

Interpretation:

- **Explained Gap:** Case differences
- **Unexplained Gap:** Potential bias or unequal treatment

Outcome: Understand which part of the disparity is due to characteristics vs. treatment.

Empirical Specification and Outcome Definition

Two-part model:

- Life sentences modeled separately as a binary outcome.
- Continuous sentencing severity normalized for non-life cases.

Normalized log sentencing ratio:

$$y_{i,c} = \frac{\log(\text{Sentence}_{i,c}) - \log(\text{Min}_{i,c})}{\log(\text{Max}_{i,c}) - \log(\text{Min}_{i,c})}$$

- Standardizes sentence length relative to legal bounds.
- Life sentences excluded from this continuous measure.

Life sentence indicator:

$$\text{Life}_{i,c} = \begin{cases} 1 & \text{if life sentence imposed} \\ 0 & \text{otherwise} \end{cases}$$

Estimation Strategy

(1) Life Sentencing (Probit Model):

$$\Pr(\text{Life}_{i,c} = 1) = \Phi(\alpha_0^c + \alpha_1^c S_i + \alpha_2^{c\top} X_{i,c})$$

Accounts for the binary nature and qualitative difference of life sentences.

(2) Non-Life Sentencing (OLS Model):

$$y_{i,c} = \beta_0^c + \beta_1^c S_i + \beta_2^{c\top} X_{i,c} + \epsilon_{i,c}$$

Estimates sex impact on normalized sentence length controlling for case factors.

Covariates and Decomposition

Covariates:

- Defendant demographics, legal representation, plea status.
- Charge degree, location, victim information.

Oaxaca–Blinder Decomposition:

$$\bar{y}_M - \bar{y}_F = (\bar{X}_M - \bar{X}_F)\hat{\beta}_M + \bar{X}_F(\hat{\beta}_M - \hat{\beta}_F)$$

- Quantifies explained vs unexplained sentencing gaps.
- Also applied within subgroups (e.g., charge type) to detect heterogeneity.

Scalability

- The framework generalizes to other offense types by updating parameters.
- Automates disparity detection across criminal code using standardized inputs.
- Designed for scalable, reproducible judicial bias analysis.

Data Sets and Characteristics

The data covers hundreds of thousands of Florida cases with detailed profiles and court info.

Data Sources:

- **Arrest/Booking Reports (County Detentions Office)**
  Includes arrest details: dates, charges, county, statute descriptions.
- **Case Reports (State Attorneys)**
  Trial details: case dates, charges, final decisions, victim info.
- **Case Reports (Clerks of Court)**
  Court processing: sentencing dates, counsel type, fines, and sentencing specifics.

Data Preparation

Challenge: No unique common identifiers across datasets due to department-specific IDs.

Solution: Fuzzy Joining

- Merge datasets by approximate matches on case attributes: statute, race, sex, charge, degree, location, etc.
- Use temporal ordering of dates (arrest $<$ booking $<$ disposition $<$ sentencing) with time constraints (e.g., booking within 3 days of arrest).

This approach helps integrate heterogeneous judicial data into a unified analytic sample.

Data Sources

Primary Data Source:

- Florida Department of Law Enforcement (2025). *Criminal Justice Data Transparency (CJDT).* `https://www.fdle.state.fl.us/CJAB/CJDT`

The CJDT framework provides comprehensive access to arrest, trial, and court records critical for this analysis.

Alesina, Alberto e Eliana La Ferrara. 2014. "A Test of Racial Bias in Capital Sentencing". *The American Economic Review* 104 (11): 3397–3433.

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